Effect of spatial dimension on the growth of particle numbers in a supercritical branching random walk

Elena Yarovaya

elena.yarovaya@math.msu.ru
Department of Probability Theory
Mechanical and Mathematical Faculty
Lomonosov Moscow State University

STOCHASTICS, ARTIFICIAL INTELLIGENCE AND MATHEMATICAL FINANCE

12-14 OCTOBER 2026, HSE, MOSCOW, RUSSIA

The modeling of the evolution of populations of particles, cells, or individuals in physics, biology, genetics, and demography has served as a stimulus for the development of stochastic processes.

In connection with this, it became necessary to introduce new classes of stochastic processes that enable the modeling and analysis of populations with complex structure and dynamics. One such class is *branching random walks*. Branching random walks are stochastic processes that combine two types of behavior:

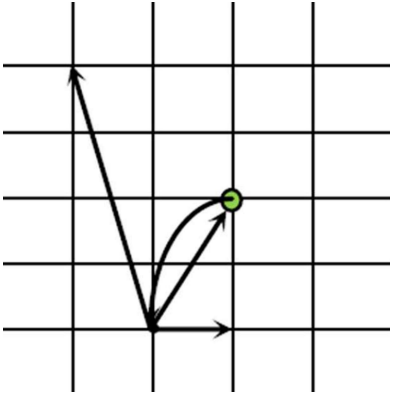
- random walks — particles move in space according to some random law;
- branching processes — particles can reproduce multiple offspring or die.

Thus, a branching random walk describes a population of particles that both move randomly and undergo reproduction events.

Some statistics on ArXiv publications: date 26.06; 23.08, and 11.10 (2026):

- Random walks — 13451; 13620; and 13884.
- Branching processes — 9370; 9609; and 9863.
- Branching random walks — 878; 893; and 914.

A branching random walk on the lattice $\mathbb{Z}^2$ with one source of branching (the green point) and one particle (the black bold point) at $t = 0$



- We consider a symmetric continuous-time branching random walk (BRW) on $\mathbb{Z}^d$, $d \in \mathbb{N}$, with a single branching source.
- **Goal**: to establish conditions for the "inheritability" of underlying random walk properties when moving from $\mathbb{Z}^d$ to $\mathbb{Z}^{d+1}$.
- **Main Result:** using the "inheritability" condition, a limit theorem is proved on the representation of the positive eigenvalue of the evolution operator of mean particle numbers in terms of the process parameters and the dimension d as the branching source intensity $\beta \to \infty$ for an arbitrary symmetric random walk.

- In classical BRW models, the dimension d is fixed.

  When comparing the exponential growth rate λ_0 of particle numbers in a BRW on lattices of different integer dimensions for a fixed source intensity β, the following **question** arises:

- *Is it possible to construct a family of random walks on $\mathbb{Z}^d$, $d \in \mathbb{N}$, such that the model for $\mathbb{Z}^{d+1}$ "inherits" the properties of the model for $\mathbb{Z}^d$?*

- To answer this, we needed to introduce the notion of **"random walk on nested lattices"** and study the influence of the dimension d on the growth of particle numbers.

Random walk on a lattice of fixed dimension

Recall:

A random walk on the lattice $\mathbb{Z}^d$ is determined by a matrix of transition intensities $A_d = (a_d(x,y))_{x,y\in\mathbb{Z}^d}$, whose elements satisfy the following conditions (axioms):

1. $a_d(x,y) \geq 0$ for $x \neq y$, $a_d(x,x) < 0$.
2. $\sum_y a_d(x,y) = 0$.
3. Symmetry: $a_d(x,y) = a_d(y,x)$.
4. Irreducibility: all points $y \in \mathbb{Z}^d$ are reachable.
5. *Finite variance:* $\sum_x |x|^2 a_d(0,x) < \infty$.
6. Spatial homogeneity: $a_d(x,y) = a_d(0,y-x) := a_d(y-x)$.

Transition probabilities of a random walk

Let $p_d(t,x,y)$ denote the *transition probability* of the random walk, i.e. the probability that at time $t \geqslant 0$ the particle is at point y given that at time $t=0$ it was at point x. Then, as $h \downarrow 0$, the following representation holds:

$$p_d(h,x,y) = \begin{cases} a_d(x,y)h + o(h) & \text{for } x \neq y, \\ 1 + a_d(x,x)h + o(h) & \text{otherwise.} \end{cases}$$

Next, without loss of generality, we may assume that at time $t=0$ the process started from the origin $x=0$.
The probability $p_d(t,x,y)$ satisfies the system of *Kolmogorov backward equations*

$$\frac{\partial p_d(t,x,y)}{\partial t} = (\mathcal{A}_d p_d(t,\cdot,y))(x), \qquad p_d(0,x,y) = \delta_y(x), \tag{1}$$

where $\mathcal{A}_d$ is a linear bounded operator on $l^p(\mathbb{Z}^d)$:

$$(\mathcal{A}_d u)(z) := \sum_{z' \in \mathbb{Z}^d} a_d(z-z')u(z'). \tag{2}$$

The quantity

$$G_\lambda(x,y) = \lim_{t\to\infty} G_\lambda(t,x,y) = \lim_{t\to\infty} \int_0^t e^{-\lambda u} p(u,x,y)\, du,$$

which is the limiting value at infinity of the Laplace transform of the transition probability $p(t,x,y)$, is called the *Green's function* of the stochastic walk.

Set $\lambda = 0$. Then, for any $t > 0$, the equality

$$G_0(t,0,0) = \int_0^t p(s,0,0)\, ds$$

holds.
The function $G_0(t,0,0)$ has a simple probabilistic interpretation: it represents the expected time spent by the particle at the point $y = 0$ up to time t, given that at the initial time $t = 0$ the particle was at the point $x = 0$.

The process of random walk on $\mathbb{Z}^d$. A simple symmetric random walk on $\mathbb{Z}^d$ is defined by the difference Laplacian $\varkappa\Delta$, where $\varkappa > 0$,

$$(\varkappa\Delta u)(z) := \varkappa \sum_{z':|z'-z|=1} (u(z') - u(z)), \qquad z \in \mathbb{Z}^d$$

and satisfies all the above conditions. The operator $\varkappa\Delta$ acts in the space $l^2(\mathbb{Z}^d)$, i.e., it is defined and bounded for $u \in l^2(\mathbb{Z}^d)$.

Branching process at a point $x \in \mathbb{Z}^d$. Branching at the source is described by the generating function $f(u) = \sum b_n u^n$, and $\beta = f'(1) = \sum b_n n$ is called the *intensity* of the branching source.

One of the main objects of study is the number of particles $\mu_{t,x}(y)$ at point $y \in \mathbb{Z}^d$ under the condition $\mu_{0,x}(y) = \delta(y - x)$.

An example of a simple symmetric BRW on $\mathbb{Z}$ with one branching source

Let us recall the notation: $\beta = \sum_n nb_n$ и $\beta_2 = \sum_n n(n-1)b_n$.

Theorem 1

If $d = 1$, $\beta > 0$, $\beta_2 < \infty$ *and* $\varkappa > 0$, *then*

$$\lim_{t\to\infty} \sup_{x\in\mathbb{Z}^d} \mathsf{E}\big(e^{-\lambda_0 t}\mu_{t,x}(y) - \lambda_0 G_{\lambda_0}(0,y)\xi_x\big)^2 = 0,$$

where

$$\lambda_0 = \sqrt{\beta^2 + 4\varkappa^2} - 2\varkappa$$

and ξ_x *is a non-degenerate random variable, for which* $\mathsf{E}\xi_x$ *and* $\mathsf{E}\xi_x^2$ *can be computed explicitly.*

Corollary

$$\lambda_0(1,\beta,\varkappa) = \beta - 2\varkappa + O(\beta^{-1}) \quad \textit{for} \quad \beta\to\infty.$$

An example of a simple symmetric BRW in $\mathbb{Z}^d$ with one branching source (Gusarov and Ya., TVP, 2026)

The listed axioms for the random walk underlying the BRW on $\mathbb{Z}^d$ are sufficient to prove the following theorems:

Theorem 2

For every $d \in \mathbb{N}$ and $\varkappa > 0$ the following representation holds:

$$\lambda_0(d, \beta, \varkappa) = \beta - 2\varkappa d + O(\beta^{-1}) \quad \textit{for} \quad \beta \to \infty.$$

Corollary

For $\beta \to \infty$ and fixed $\varkappa > 0$, for each $d \in \mathbb{N}$, the follwing relation holds

$$\lim_{\beta \to \infty} (\lambda_0(1, \beta, \varkappa) - \lambda_0(d, \beta, \varkappa)) = 2\varkappa(d-1).$$

Random walks on nested lattices

To reconcile the properties of random walks when moving from $\mathbf{Z}^d$ to $\mathbf{Z}^{d+1}$, for an arbitrary symmetric random walk with generator $A_d = (a_d(x, y))_{x,y\in\mathbb{Z}^d}$, we introduce an additional condition to the standard set of axioms given above.

Remark

In contrast to a random (stochastic) walk on a fixed-dimensional lattice, for sets of random walks on lattices of different dimensions that include different step sizes simultaneously or sequentially (e.g., small steps + large "jumps") or walks, we will use the term **random (stochastic) walks on nested lattices**.

We use "random walks on nested lattices" for the underlying process that allows comparison between $\mathbb{Z}^d$ and $\mathbb{Z}^{d+1}$.

Hereafter, the function $a_d(0,x)$ for the sake of brevity will be denoted by omitting the first zero argument: $a_d(0,x) := a_d(x)$. For $a_d(x)$ it is sometimes convenient to use coordinate notation

$$a_d(x) \equiv a_d(\underbrace{x_1, x_2, \ldots, x_d}_{d \text{ arguments}}), \qquad x = (x_1, x_2, \ldots, x_d) \in \mathbb{Z}^d.$$

Definition 1

We say that a collection of random walks with intensity matrices $A_d = (a_d(x,y))_{x,y\in\mathbb{Z}^d}$, $d = 1, 2, \ldots$, satisfying Axioms 1–6, forms **random walks on nested lattices** if for every $d = 1, 2, \ldots$ the following identity holds:

$$a_{d+1}(x_1, \ldots, x_d, 0) \equiv a_d(x_1, , \ldots, x_d), \qquad x = (x_1, , \ldots, x_d) \neq 0. \qquad (*)$$

Remark

Condition $(*)$ is obviously satisfied in any dimension d for the Laplace operator Δ and, in fact, arose as a generalization of a similar condition for Δ.

The notion of a random walks on nested lattices can be extended.

Definition 2

We say that a collection of random walks with intensity matrices $A_d = (a_d(x, y))_{x,y\in\mathbb{Z}^d}$, $d = 1, 2, \ldots$, satisfying axioms 1–6, forms **increasing random walks on nested lattices** if for every $d = 1, 2, \ldots$ the following relation holds:

$$a_{d+1}(x_1, \ldots, x_d, 0) \geq a_d(x_1, \ldots, x_d), \qquad x = (x_1, , \ldots, x_d) \neq 0.$$

Remark

The difference in the definitions is that in the second definition the two parts of the relationship between a_{d+1} and a_d are related by inequality, while in the first definition the two parts of the relationship are related by identity (equality).

Next we need the Fourier transform which, for transition intensities, has the form:

$$\phi_d(\theta) = \sum_{x \in \mathbb{Z}^d} a_d(x) e^{i\langle x, \theta \rangle}, \quad \theta \in [-\pi, \pi]^d.$$

$\phi_d(0) = 0$, $\phi_d(\theta) < 0$ for $\theta \neq 0$.

The following two lemmas are key in the study of the properties of random walks on nested lattices.

Lemma 1

For increasing random walks on nested lattices, for each d the following inequality holds

$$|a_{d+1}(0)| \geq |a_d(0)|.$$

Lemma 2

The Fourier transform

$$\phi_d(\theta) = \sum_{x \in \mathbb{Z}^d} a_d(x) e^{i\langle x, \theta \rangle}$$

decreases monotonically as the lattice dimension d grows:

$$\phi_{d+1}(\theta_1, \ldots, \theta_d, \theta_{d+1}) \leq \phi_d(\theta_1, \ldots, \theta_d).$$

An important characteristic of a random walk is its Green's function:

$$G_\lambda(d,x) = \int_0^\infty e^{-\lambda u} p(u,x,y)du = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{e^{i\langle x,\theta\rangle}}{\lambda - \phi_d(\theta)}\, d\theta.$$

In particular, a random walk is *recurrent* if $G_0(d,0) = \infty$, and *non-recurrent* if $G_0(d,0) < \infty$.

- With finite variance of random walk jumps there is recurrence for $d = 1, 2$ and transience for $d \geq 3$.
- With heavy tails (power law) for $|x| \to \infty$:

$$a_d(x) \sim \frac{H(x/|x|)}{|x|^{d+\alpha}} \Rightarrow \sum_x |x|^2 a_d(0,x) = \infty$$

recurrence is possible only for $d = 1$ and $\alpha \in [1,2)$ (Axiom 5 is not satisfied, but the main approaches to the analysis remain valid).

- Branching at the source is described by the generating function $f(u) = \sum b_n u^n$ with intensity $\beta = f'(1)$.
- The evolution operator of BRW has the form $\mathcal{H}_\beta = \mathcal{A}_d + \beta\mathcal{V}_0$, where $\mathcal{V}_0$ is a potential concentrated at zero.
- The Green function $G_\lambda(d, 0)$ of $\mathcal{H}_\beta$ determines the critical intensity β_c:

$$\beta_c = \begin{cases} 0, & \text{if } G_0(d,0) = \infty \text{ (recurrent random walk)}, \\ G_0^{-1}(d,0), & \text{if } G_0(d,0) < \infty \text{ (transient random walk)}. \end{cases}$$

- For $\beta > \beta_c$ the process is in the supercritical regime and the particle numbers at every lattice point grow exponentially.

- For a finite variance: $\beta_c = 0$ for $d = 1, 2$; $\beta_c > 0$ for $d \geq 3$.
- For infinite variance (power-law tails $\approx |z|^{-(d+\alpha)}$, $\alpha \in (0, 2)$):
  - $1/\beta_c = G_0 < \infty$ is possible already for $d = 1$ ($\alpha < 1$) and for $d \geq 2$ ($\alpha < 2$).
  - Consequently, in contrast to the case of a finite variance of jumps, $\beta_c > 0$ may occur even on $\mathbb{Z}$ and $\mathbb{Z}^2$.
- Thus, even with supercritical branching at the source, a branching random walk may remain subcritical due to the transience of the underlying random walk.

The existence condition for $\lambda_0(d,\beta) > 0$

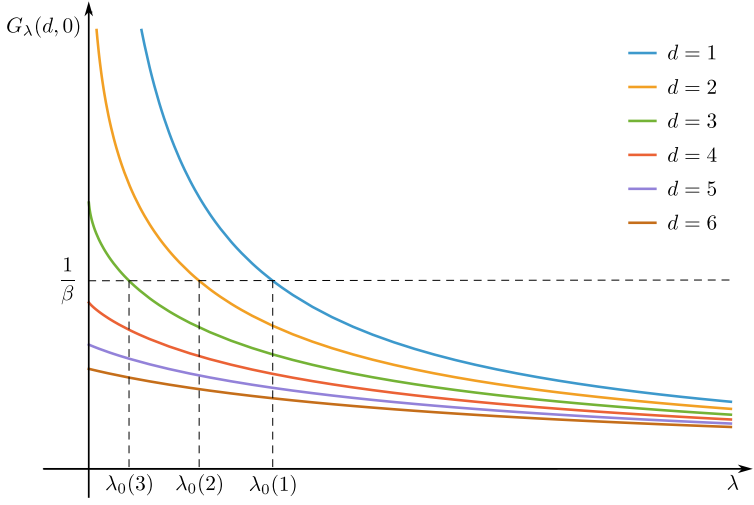
It is known that in a supercritical BRW with branching intensity $\beta > 0$, a **positive eigenvalue** $\lambda_0(d,\beta)$ of the operator $\mathcal{H}_\beta = \mathcal{A}_d + \beta\mathcal{V}_0$ arises as the solution of the equation

$$G_\lambda(d,0) = \frac{1}{\beta}, \qquad \lambda > 0.$$

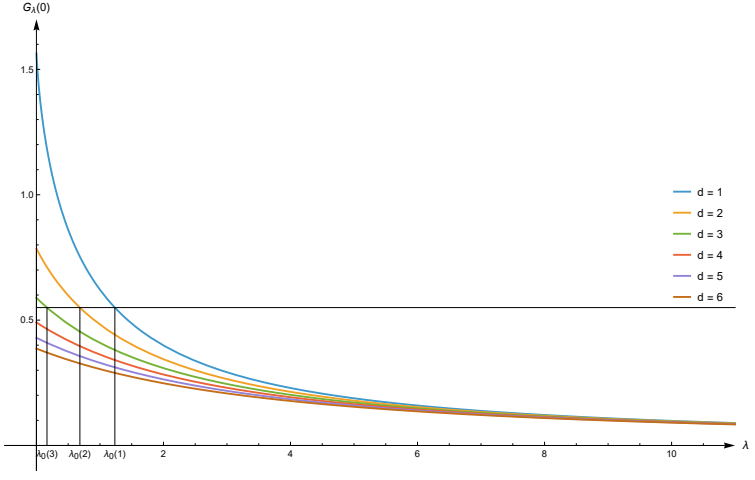
For a BRW with a single source of particle branching, having branching intensity $\beta > 0$ and located at the origin on $\mathbb{Z}^d$, this equation has a unique solution $\lambda_0(d) := \lambda_0(d,\beta) > 0$ provided $\beta > \beta_c$, where

$$\beta_c = \frac{1}{G_0(d,0)}.$$

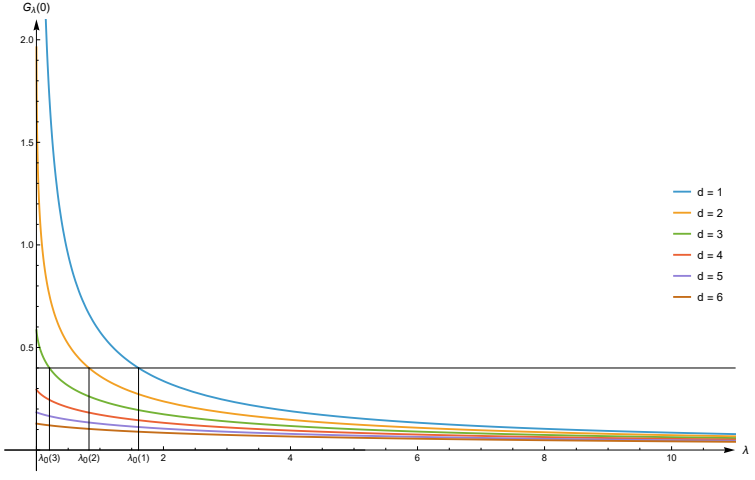
Behavior of the solution $\lambda_0(d)$ of the equation $1/\beta = G_\lambda(0)$ depending on the lattice dimension d. The case of a finite variance of random walk jumps



Behavior of the solution $\lambda_0(d)$ of the equation $1/\beta = G_\lambda(0)$ depending on the lattice dimension d. The case of a heavy-tailed random walk jump distribution (infinite variance) for $\alpha \in (0, 1)$



Behavior of the solution $\lambda_0(d)$ of the equation $1/\beta = G_\lambda(0)$ depending on the lattice dimension d. The case of a heavy-tailed random walk jump distribution (infinite variance) for $\alpha \in [1, 2)$



We consider the equation

$$G_\lambda(d,0) = \frac{1}{\beta}. \qquad (*)$$

Let n be the maximal lattice dimension for which a positive solution $\lambda_0(d,\beta)$ exists.

Theorem 3

For $d = 1,\dots,n$ and $\beta > G_0(d,0)^{-1}$ the equation $G_\lambda(d,0) = \beta^{-1}$ has a unique solution $\lambda_0(d,\beta) \in (0,\beta)$. Moreover, $\lambda_0(d,\beta)$ strictly decreases with d:

$$0 < \lambda_0(n,\beta) < \cdots < \lambda_0(d+1,\beta) < \lambda_0(d,\beta) < \cdots < \lambda_0(1,\beta) < \beta.$$

Proof idea: we first show monotonic decrease of $G_\lambda(d,0)$ in d based Lemma 2 given above and integration over all θ_{d+1}, and then prove monotonicity of the inverse function.

Corollary 1. Dimensions $d = 1, 2$ (the case of a finite variance of random walk jumps)

In dimensions $d = 1$ and $d = 2$, the function $G_\lambda(d, 0)$ diverges as $\lambda \to 0$, so the equation $G_\lambda = \beta^{-1}$ has a positive solution **for any** $\beta > 0$.

Moreover,

$$\lim_{\beta \to 0} \big(\lambda_0(1, \beta) - \lambda_0(2, \beta)\big) = 0.$$

Physical meaning: for small branching intensity, the difference between $\mathbb{Z}$ and $\mathbb{Z}^2$ disappears.

Corollary 2. Dimensions $d = 1$ and $\alpha \in (0, 2)$ (the case of infinite variance of random walk jumps)

- For dimension $d = 1$, for each fixed $\alpha \in [1, 2)$, the function $G_\lambda(d, 0)$ diverges as $\lambda \to 0$, and so the equation $G_\lambda = \beta^{-1}$ has a positive solution **for all** $\beta > 0$. In the remaining cases, a solution exists when $G_0(d, 0) > \beta^{-1}$.

- For dimension $d = 1$, for each fixed $\alpha \in (0, 1)$ and $d \geq 2$ the solution exists when $G_0(d, 0) > \beta^{-1}$.

The main result of the work is the following asymptotic formula for the positive eigenvalue $\lambda_0(d, \beta)$ of the evolutionary operator of mean number of particles in a supercritical branching random walk, which holds for a wide class of transition intensities, including models where the random walk jumps have infinite variance.

Theorem 4

If $\beta \to \infty$, then for each $d \in \mathbb{N}$ we have

$$\lambda_0(d, \beta) = \beta + a_d(0) + O(\beta^{-1}),$$

where $a_d(0) < 0$ is the diagonal element of the matrix of transition intensities $A_d = (a_d(x, y))_{x,y \in \mathbb{Z}^d}$.

Proof method: based on the implicit function theorem and a new condition for increasing random walks on nested lattices.

Difference of λ_0 for different d under "strong" branching

Theorem 4 directly implies:

Corollary

$$\lim_{\beta \to \infty} (\lambda_0(1, \beta) - \lambda_0(d, \beta)) = a_1(0) - a_d(0).$$

Remark

The value of $a_d(0)$ depends on d, so the limit difference may be nonzero. This reflects the influence of dimension even for very strong branching.

- A construction of random walks on nested lattices is developed, allowing a natural transition from $\mathbb{Z}^d$ to $\mathbb{Z}^{d+1}$.
- The **monotone dependence** of the eigenvalue $\lambda_0(d, \beta)$ on d is proved.
- The **asymptotic formula** $\lambda_0 = \beta + a_d(0) + O(1/\beta)$ as $\beta \to \infty$ is obtained.
- It is shown that for recurrent random walks (case of finite variance of random walk jumps: $d = 1, 2$) λ_0 exists for any $\beta > 0$, and the difference $\lambda_0(1, \beta) - \lambda_0(2, \beta)$ tends to zero as $\beta \to 0$.
- It is established that in the presence of a **heavy tailed** random walk jump distribution the critical value β_c can be positive even in $d = 1$ and $d = 2$, which is qualitatively different from the finite variance case. However, the methods of proof based on the idea of "inheritance" remain.
- The results are important for understanding "threshold phenomena" in branching random walks.

- E. B. Yarovaya, A. S. Gusarov (2026). Estimation of particle number growth in a supercritical branching random walk by the method of successive approximations. *Theory Probab. Appl.*, 71:3, 473-495
- A. Rytova, E. Yarovaya (2021). Heavy-tailed branching random walks on multidimensional lattices. A moment approach. *Proceedings of the Royal Society of Edinburgh: Section A Mathematics*, 151(3), 971–992
- E. B. Yarovaya (2025). *Branching random walks in non homogeneos environment*, MCCME (Moscow Center for Continuous Mathematical Education), Moscow
- S. A. Molchanov, E. B. Yarovaya (2012), Limit theorems for the Green function of the lattice Laplacian under large deviations of the random walk, Izv. Math., 76:6, 1190–1217

THANKS FOR YOUR ATTENTION